API DOCUMENTATION

AI-Powered Cybersecurity Incident Response Assistant

REST and Server-Sent Events API of the FastAPI backend: accounts, assistant, feedback, machine-learning models and incident search

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1. Overview

Property	Value
Base URL	`http://localhost:8000` (start with `python scripts/api/run_api.py`)
Format	JSON requests and responses; `/chat/stream` returns Server-Sent Events; one endpoint takes a CSV upload
Interactive docs	Swagger UI at `/docs`, OpenAPI schema at `/openapi.json`
Authentication	`Authorization: Bearer <token>` from `/auth/signup` or `/auth/login`; needed for the assistant, conversation and feedback-writing endpoints (🔒)
Request IDs	Every response has an `X-Request-ID` header (taken from the request or generated)
CORS	Only the Streamlit origin (`localhost:8501`) by default; `API_CORS_ORIGINS` changes it

All samples in this document are **real responses** of the API, recorded on 11 October 2026 with the assistant running `nvidia:openai/gpt-oss-20b` and the project's trained models and search indexes. Long lists and texts are shortened (`… N more`, `…`).

1.1 Endpoints

Method	Path	Purpose	Auth
`GET`	`/health`	Service status	
`POST`	`/auth/signup`	Create an account	
`POST`	`/auth/login`	Sign in	
`POST`	`/auth/logout`	Sign out	🔒
`GET`	`/auth/me`	Current user	🔒
`POST`	`/chat`	Ask the assistant	🔒
`POST`	`/chat/stream`	Ask the assistant (live trace)	🔒
`GET`	`/chat/conversations`	Your conversations	🔒
`GET`	`/chat/{thread_id}/history`	Conversation history	🔒
`PATCH`	`/chat/{thread_id}`	Rename a conversation	🔒
`DELETE`	`/chat/{thread_id}`	Delete a conversation	🔒
`POST`	`/judge`	Grade an answer (LLM-as-Judge)	🔒
`POST`	`/feedback`	Rate an answer	🔒
`GET`	`/feedback`	List feedback	
`GET`	`/feedback/stats`	Feedback statistics	
`GET`	`/feedback/export.csv`	Export label corrections	

Method	Path	Purpose	Auth
`POST`	`/ml/classify`	Attack type of a flow	
`POST`	`/ml/risk`	Intrusion risk and anomaly of a flow	
`POST`	`/ml/explain`	Full analysis with SHAP reasons	
`POST`	`/ml/prioritize`	Priority of one flow	
`POST`	`/ml/prioritize/batch`	Rank many flows / incidents	
`POST`	`/ml/prioritize/csv`	Rank flows from a CSV file	
`POST`	`/search`	Find similar historical incidents	
`GET`	`/incidents/{incident_id}`	Look up a historical incident	
`GET`	`/attack-types`	List attack types	
`GET`	`/attack-types/{name}`	Profile of an attack type	

🔒 = bearer token required.

1.2 Status codes

Code	When
`200` / `201` / `204`	Success (`201` for sign-up and new feedback, `204` for logout and delete)
`200` with `"status": "needs_follow_up"`	ML endpoints: key metrics (duration, packets, bytes) are missing; the response says which
`200` with `"route": "blocked"`	Assistant: the input guard blocked the message; fixed reply
`401`	Missing, expired or revoked token; wrong username or password
`404`	Unknown incident, attack type or conversation, or another user's conversation
`409`	Username already taken
`413`	CSV larger than 5 MB
`422`	Invalid input (validation details in `detail`) or impossible flow values
`429`	Too many failed logins for a username (`Retry-After` header)
`502`	The LLM provider returned an error (quota, billing, outage); `detail` has its message
`503`	Assistant not configured (no LLM key), or the search indexes failed to load
`500`	Unexpected error; the body has a `request_id` to find it in the server log

Errors have the shape `{"detail": "..."}` (validation errors: a list of `{loc, msg, type}`).

1.3 Flow input

The `/ml` endpoints and the assistant accept a network flow as **any subset of the 24 metrics** below, or an `incident_id` of a historical incident. Values can be numbers or strings with units (`"2 s"` , `"1.5 KB"` , `"300`

`ms"`). Missing metrics are derived where possible, otherwise filled with normal-traffic medians and reported in `input_quality`; if duration, packets or bytes are all missing, the response asks for them (`needs_follow_up`). Impossible values are rejected with `422`.

Field	Type	Required	Description
`flow_duration`	number or string	no	Total flow duration in microseconds (e.g. '2 s' or 2000000).
`total_fwd_packets`	number or string	no	Number of packets sent from source to destination.
`total_bwd_packets`	number or string	no	Number of packets sent from destination back to source.
`total_len_fwd`	number or string	no	Total bytes sent forward (source to destination).
`total_len_bwd`	number or string	no	Total bytes sent backward (destination to source).
`flow_packets_s`	number or string	no	Packets per second over the whole flow.
`fwd_packets_s`	number or string	no	Forward packets per second.
`bwd_packets_s`	number or string	no	Backward packets per second.
`fwd_packet_length_mean`	number or string	no	Mean size of forward packets in bytes.
`bwd_packet_length_mean`	number or string	no	Mean size of backward packets in bytes.
`min_packet_length`	number or string	no	Smallest packet size in bytes.
`max_packet_length`	number or string	no	Largest packet size in bytes.
`packet_length_mean`	number or string	no	Mean packet size in bytes (both directions).
`packet_length_std`	number or string	no	Standard deviation of packet size in bytes.
`syn_flag_count`	number or string	no	1 if the TCP SYN flag was seen, else 0.
`ack_flag_count`	number or string	no	1 if the TCP ACK flag was seen, else 0.
`rst_flag_count`	number or string	no	1 if the TCP RST flag was seen, else 0.
`psh_flag_count`	number or string	no	1 if the TCP PSH flag was seen, else 0.

Field	Type	Required	Description
`fin_flag_count`	number or string	no	1 if the TCP FIN flag was seen, else 0.
`down_up_ratio`	number or string	no	Backward packets divided by forward packets (rounded down).
`init_win_bytes_forward`	number or string	no	Initial TCP window size in the forward direction, bytes (-1 if not set).
`init_win_bytes_backward`	number or string	no	Initial TCP window size in the backward direction, bytes (-1 if not set).
`active_mean`	number or string	no	Mean time the flow was active before going idle, microseconds.
`idle_mean`	number or string	no	Mean time the flow was idle before becoming active, microseconds.
`incident_id`	string	no	ID of a historical incident, e.g. 'INC-000123'. When given, the incident's stored metrics are used and any metric fields are ignored.

2. System

GET /health

Service status

Whether the ML models, the assistant (LLM) and the search indexes are ready.

`status` is "ok" when the models and the assistant are ready and the indexes are not in error; otherwise "degraded" (the direct ML endpoints may still work).

Property	Value
Method	`GET`
Path	`/health`
Authentication	None

Request format

No request body.

Response format

`200` · application/json

Field	Type	Required	Description
`status`	`ok` | `degraded`	yes	
`ml_models_loaded`	boolean	yes	
`attack_types`	integer	yes	
`assistant_ready`	boolean	yes	
`assistant_error`	string	no	
`llm_model`	string	no	
`classifier_version`	string	no	"original" or a retrained version.
`search_index`	`not_loaded` | `loading` | `ready` | `error`	yes	
`search_index_error`	string	no	
`versions`	object	yes	

Sample request

```
GET /health HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
```

```
  "ml_models_loaded": true,
  "attack_types": 15,
  "assistant_ready": true,
  "assistant_error": null,
  "llm_model": "nvidia:openai/gpt-oss-20b",
  "classifier_version": "original",
  "search_index": "ready",
  "search_index_error": null,
  "versions": {
    "fastapi": "0.143.0",
    "langgraph": "1.2.14",
    "langchain-anthropic": "1.7.5",
    "scikit-learn": "1.7.2",
    "shap": "0.48.0"
  }
}
```

3. Accounts

POST /auth/signup

Create an account

Create an account and sign in. Username: 3-32 letters, digits, '.', '_' or '-' (not case-sensitive). Password: 8-128 characters.

Property	Value
Method	`POST`
Path	`/auth/signup`
Authentication	None

Request format

Field	Type	Required	Description
`username`	string	yes	(min length 1, max length 64)
`password`	string	yes	(min length 1, max length 256)

Response format

`201` · application/json

Field	Type	Required	Description
`token`	string	yes	Send as 'Authorization: Bearer <token>'.
`expires_at`	string	yes	
`user`	object	yes	

Sample request

```
POST /auth/signup HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "username": "analyst_demo",
  "password": "demo-passphrase-2026"
}
```

Sample response

```
HTTP/1.1 201

{
  "token": "<token>",
  "expires_at": "2026-10-18T09:29:00.440264+00:00",
  "user": {
    "id": 1,
    "username": "analyst_demo",
    "created_at": "2026-10-11T09:29:00.430735+00:00"
```

```
  }
}
```

POST /auth/login

Sign in

Property	Value
Method	`POST`
Path	`/auth/login`
Authentication	None

Request format

Field	Type	Required	Description
`username`	string	yes	(min length 1, max length 64)
`password`	string	yes	(min length 1, max length 256)

Response format

`200` · application/json

Field	Type	Required	Description
`token`	string	yes	Send as 'Authorization: Bearer <token>'.
`expires_at`	string	yes	
`user`	object	yes	

Sample request

```
POST /auth/login HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "username": "analyst_demo",
  "password": "demo-passphrase-2026"
}
```

Sample response

```
HTTP/1.1 200

{
  "token": "<token>",
  "expires_at": "2026-10-18T09:29:00.507677+00:00",
  "user": {
    "id": 1,
    "username": "analyst_demo",
    "created_at": "2026-10-11T09:29:00.430735+00:00"
  }
}
```

POST /auth/logout

Sign out

Revoke the token. Always succeeds (logging out twice is harmless).

Property	Value
Method	`POST`
Path	`/auth/logout`
Authentication	Bearer token (the session to end)

Request format

No request body.

Response format

`204` · no body

Sample request

```
POST /auth/logout HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
```

Sample response

```
HTTP/1.1 204

(no body)
```

GET /auth/me

Current user

Property	Value
Method	`GET`
Path	`/auth/me`
Authentication	Bearer token required

Request format

No request body.

Response format

`200` · application/json

Field	Type	Required	Description
`id`	integer	yes	
`username`	string	yes	
`created_at`	string	yes	

Sample request

```
GET /auth/me HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
```

Sample response

```
HTTP/1.1 200

{
  "id": 1,
  "username": "analyst_demo",
  "created_at": "2026-10-11T09:29:00.430735+00:00"
}
```

4. Assistant

POST /chat

Ask the assistant

Answer one message. The response includes the agent trace and guardrail results.

Pass the returned `thread_id` with the next message to continue the conversation (follow-up answers, "what about its risk?").

Property	Value
Method	`POST`
Path	`/chat`
Authentication	Bearer token required

Request format

Field	Type	Required	Description
`message`	string	yes	The user's question. (min length 1, max length 4000)
`thread_id`	string	no	Conversation ID from an earlier response. Omit to start a new conversation.
`judge`	boolean	no	Grade the answer with LLM-as-Judge (one extra LLM call). Omit to use JUDGE_MODE from .env.

Response format

`200` · application/json

Field	Type	Required	Description
`thread_id`	string	yes	
`answer`	string	yes	
`route`	string	yes	Supervisor route, or 'blocked' when the input guardrail stopped it.
`route_reason`	string	no	
`agents_run`	array of string	no	
`needs_follow_up`	boolean	yes	True when the answer is a question asking for missing metrics.
`findings`	object	no	Each agent's findings, used for the final answer.
`evidence`	array of object	no	Every tool call this turn: agent, tool, args, result.
`trace`	array of object	no	Guardrail, route, agent, tool and final-answer events.
`guardrails`	object	no	

Field	Type	Required	Description
`turn_id`	string	no	ID of this turn; send it with POST /feedback.
`judgement`	object	no	LLM-as-Judge result when requested: verdict, overall, scores, reasons, unsupported claims.
`elapsed_ms`	integer	yes	

Sample request

```
POST /chat HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
Content-Type: application/json

{
  "message": "What attack is incident INC-004393?"
}
```

Sample response

```
HTTP/1.1 200

{
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
  "answer": "**Assessment**\n\nIncident INC-004393 has been classified as **Infiltration** with 90 %
prediction confidence. The historical database also labels it as Infiltration, providing strong
agreement.\n\n**Why this label?**\n\n- Very long, low-and-slow flow (≈ 68 s) with only 40 packets, indicating
a stealth reconnaissance or C&C channel.  \n- Numerous idle periods (≈ 33 s each) and tiny packets (average ≈
95 B) are typical of low-bandwidth intrusion attempts.  \n- Only a SYN/ACK handshake is visible; no data
exfiltration or reset packets.  \n- Port 444 is commonly used for custom communication channels by threat
actors.  \n\n**Similar past incidents**\n\n- No prior incident IDs are cited in the findings; the …",
  "route": "classify",
  "route_reason": "User asks to identify the attack type of a specific incident.",
  "agents_run": [
    "classification"
  ],
  "needs_follow_up": false,
  "findings": {
    "classification": "**Incident INC-004393**\n\n**Model Prediction**\n- **Attack type:** *Infiltration*
\n- **Confidence:** 90 % (high)  \n- **Alternative labels:** BENIGN (9 %), DoS
Slowhttptest (1 %)\n\n**Historical Label** (from the incident database) – **Infiltration** – matches the
prediction.\n\n**Key Metrics (used in the prediction)**\n- Duration: 67 917 167 µs (≈68 s) – relatively long
flow with low activity  \n- Total packets: 20 FWD + 20 BWD (≈0.59 packets/s) – sparse traffic  \n-
Forward/Backward byte size: 3 738 B / 120 B – very small overall payload  \n- Mean forward packet
size: 186.9 B; mean backward packet size: 6 B (very few tiny packets)  \n- SYN/ACK flags present (1 each), no
RST/PSH/FIN – classic TCP hand …"
  },
  "evidence": [
    {
      "agent": "classification",
      "tool": "lookup_incident",
      "args": {
        "incident_id": "INC-004393"
      },
      "result": {
        "status": "ok",
        "incident_id": "INC-004393",
        "historical_label": "Infiltration",
        "dst_port": 444,
        "input": {
          "input_quality": "full",
          "known_metrics": "24/24",
```

```
          "derived": [],
          "filled_with_normal_values": [],
          "warnings": []
        },
        "metrics": {
          "Flow duration": 67917167.0,
          "Forward packets": 20.0,
          "Backward packets": 20.0,
          "Forward bytes": 3738.0,
          "Backward bytes": 120.0,
          "Flow packet rate": 0.589,
          "Forward packet rate": 0.2945,
          "Backward packet rate": 0.2945,
          "Mean forward packet size": 186.9,
          "Mean backward packet size": 6.0,
          "Minimum packet size": 6.0,
          "Maximum packet size": 705.0,
          "Mean packet size": 94.9756,
          "Packet size std. dev.": 181.0974,
          "SYN flag": 1.0,
          "ACK flag": 1.0,
          "RST flag": 0.0,
          "PSH flag": 0.0,
          "FIN flag": 0.0,
          "Download/upload ratio": 1.0,
          "Initial TCP window (forward)": 255.0,
          "Initial TCP window (backward)": 1452.0,
          "Mean active time": 566477.5,
          "Mean idle time": 33100000.0
        }
      }
    },
    "… 1 more"
  ],
  "trace": [
    {
      "type": "guard_input",
      "action": "allowed",
      "category": null,
      "matched": null,
      "redactions": [],
      "truncated": false,
      "elapsed_ms": 0
    },
    "… 9 more"
  ],
  "guardrails": {
    "input": {
      "action": "allowed",
      "category": null,
      "matched": null,
      "redactions": [],
      "truncated": false
    },
    "output": {
      "issues": [],
      "changed": false
    }
  },
  "turn_id": "1418ee75519f4518a35e64bb48945262",
  "judgement": null,
  "elapsed_ms": 44957
}
```


POST /chat/stream

Ask the assistant (live trace)

Answer one message as Server-Sent Events.

- `event: trace` one per step (guardrail, route, agent start/end, tool result, final answer)
- `event: result` the full response (same shape as POST /chat), sent last
- `event: error` if the run fails part-way

Property	Value
Method	`POST`
Path	`/chat/stream`
Authentication	Bearer token required

Request format

Field	Type	Required	Description
`message`	string	yes	The user's question. (min length 1, max length 4000)
`thread_id`	string	no	Conversation ID from an earlier response. Omit to start a new conversation.
`judge`	boolean	no	Grade the answer with LLM-as-Judge (one extra LLM call). Omit to use JUDGE_MODE from .env.

Response format

`200` · text/event-stream

Sample request

```
POST /chat/stream HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
Content-Type: application/json

{
  "message": "And how risky is it?",
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806"
}
```

Sample response

```
HTTP/1.1 200

event: trace
data: {"type": "guard_input", "action": "allowed", "category": null, "matched": null, "redactions": [],
"truncated": false, "elapsed_ms": 0}

event: trace
data: {"type": "route", "route": "risk", "reason": "User asks about the risk level of a previously identified
incident", "plan": ["risk_priority"], "elapsed_ms": 1576}

event: trace
data: {"type": "agent_start", "agent": "risk_priority"}

event: trace
data: {"type": "tool_call", "agent": "risk_priority", "tool": "assess_intrusion_risk", "args":
{"incident_id": "INC-004393"}}
```

```
… 4 more trace events …

event: result
data: {"thread_id": "6ce55595a2714d46ba1d3a5122f2d806", "answer": "**Risk Assessment**\n\n- **Malicious-
intent probability:** **99.82 %**  \n- **Risk level:** **Critical** (P1 priority)\n\n**W …", "route": "risk",
"route_reason": "User asks about the risk level of a previously identified incident", "agents_run":
["risk_priority"], "needs_follow_up": false, "findings": {"risk_priority": "**Intrusion probability and risk
level**\n- Probability of malicious intent: **99.82 %**  \n- Risk level: **Critical**\n\n** …"}, "evidence":
[{"agent": "risk_priority", "tool": "assess_intrusion_risk", "args": …
```

GET /chat/conversations

Your conversations

Your saved conversations, most recently used first.

Property	Value
Method	`GET`
Path	`/chat/conversations`
Authentication	Bearer token required

Request format

No request body.

Response format

`200` · application/json

Sample request

```
GET /chat/conversations HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
```

Sample response

```
HTTP/1.1 200

[
  {
    "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
    "title": "What attack is incident INC-004393?",
    "created_at": "2026-10-11T09:29:00.529149+00:00",
    "updated_at": "2026-10-11T09:30:17.875300+00:00"
  }
]
```

GET /chat/{thread_id}/history

Conversation history

Property	Value
Method	`GET`
Path	`/chat/{thread_id}/history`

Property	Value
Authentication	Bearer token required
Parameters	`thread_id` (path, required)

Request format

No request body.

Response format

`200` · application/json

Field	Type	Required	Description
`thread_id`	string	yes	
`messages`	array of object	yes	

Sample request

```
GET /chat/6ce55595a2714d46ba1d3a5122f2d806/history HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
```

Sample response

```
HTTP/1.1 200

{
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
  "messages": [
    {
      "role": "user",
      "content": "What attack is incident INC-004393?"
    },
    {
      "role": "assistant",
      "content": "**Assessment**\n\nIncident INC-004393 has been classified as **Infiltration** with 90 %
prediction confidence. The historical database also labels it as Infiltration, providing strong
agreement.\n\n**Why this label?**\n\n- Very long, low-and-slow flow (≈ 68 s) with only 40 packets, indicating
a stealth reconnaissance or C&C channel.  \n- Numerous idle periods (≈ 33 s each) and tiny packets (average ≈
95 …"
    },
    "… 2 more"
  ]
}
```

PATCH /chat/{thread_id}

Rename a conversation

Property	Value
Method	`PATCH`
Path	`/chat/{thread_id}`
Authentication	Bearer token required

Property	Value
Parameters	`thread_id` (path, required)

Request format

Field	Type	Required	Description
`title`	string	yes	(min length 1, max length 200)

Response format

`200` · application/json

Field	Type	Required	Description
`thread_id`	string	yes	
`title`	string	yes	
`created_at`	string	yes	
`updated_at`	string	yes	

Sample request

```
PATCH /chat/6ce55595a2714d46ba1d3a5122f2d806 HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
Content-Type: application/json

{
  "title": "Infiltration case"
}
```

Sample response

```
HTTP/1.1 200

{
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
  "title": "Infiltration case",
  "created_at": "2026-10-11T09:29:00.529149+00:00",
  "updated_at": "2026-10-11T09:30:17.875300+00:00"
}
```

DELETE /chat/{thread_id}

Delete a conversation

Property	Value
Method	`DELETE`
Path	`/chat/{thread_id}`
Authentication	Bearer token required
Parameters	`thread_id` (path, required)

Request format

No request body.

Response format

`204` · no body

Sample request

```
DELETE /chat/6ce55595a2714d46ba1d3a5122f2d806 HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
```

Sample response

```
HTTP/1.1 204

(no body)
```

POST /judge

Grade an answer (LLM-as-Judge)

Score an answer 1-5 on faithfulness, correctness, relevance, actionability and safety, list unsupported claims, and give a verdict (pass / needs_review / fail).

Send a `thread_id` to grade that conversation's latest answer with its real evidence, or a `question` + `answer` (optionally with `findings` and `evidence`) to grade any answer. Grading a conversation needs its owner to be signed in.

Property	Value
Method	`POST`
Path	`/judge`
Authentication	Bearer token required when grading a conversation (`thread_id`)

Request format

Field	Type	Required	Description
`thread_id`	string	no	
`question`	string	no	
`answer`	string	no	
`route`	string	no	Route of the answer (actionability is scored for recommend / investigate).
`findings`	object	no	Agent findings the answer was written from.
`evidence`	array of object	no	Tool calls: agent, tool, args, result.

Response format

`200` · application/json

Field	Type	Required	Description
`verdict`	`pass` | `needs_review` | `fail` | `error`	yes	
`overall`	number	no	
`scores`	object	no	
`reasons`	object	no	
`applicable`	array of string	no	
`unsupported_claims`	array of string	no	
`error`	string	no	
`judge_model`	string	no	

Sample request

```
POST /judge HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
Content-Type: application/json

{
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806"
}
```

Sample response

```
HTTP/1.1 200

{
  "verdict": "pass",
  "overall": 5.0,
  "scores": {
    "faithfulness": 5,
    "correctness": 5,
    "relevance": 5,
    "actionability": 5,
    "safety": 5
  },
  "reasons": {
    "faithfulness": "All factual claims directly match the evidence and findings.",
    "correctness": "Risk probability, level, anomaly, and priority all match the tool output and findings.",
    "relevance": "Answer addresses risk assessment and covers key findings of the evidence.",
    "actionability": "Steps are specific, prioritized, and actionable for an analyst.",
    "safety": "No harmful instructions; uncertainties and caveats are stated."
  },
  "applicable": [
    "faithfulness",
    "correctness",
    "… 2 more"
  ],
  "unsupported_claims": [],
  "error": null,
  "judge_model": "nvidia:openai/gpt-oss-20b"
}
```

5. Feedback

POST /feedback

Rate an answer

Thumbs up/down, an optional comment and an optional corrected attack type for an answer.

The question, answer and model prediction are taken from the conversation itself. Rating the same turn again updates the earlier feedback. A corrected label is shown in later answers about the same incident and is used when the classifier is retrained. Only the owner of the conversation can rate it; the rating records their username.

Property	Value
Method	`POST`
Path	`/feedback`
Authentication	Bearer token required

Request format

Field	Type	Required	Description
`thread_id`	string	yes	
`turn_id`	string	no	Turn to rate (from the chat response). Omit to rate the conversation's latest answer.
`rating`	`up` | `down`	yes	
`comment`	string	no	
`corrected_label`	string	no	Correct attack type, if the prediction was wrong (one of the 15 types; case-insensitive).

Response format

`201` · application/json

Field	Type
`id`	number
`created_at`	string
`updated_at`	string
`thread_id`	string
`turn_id`	string
`rating`	string
`comment`	string
`corrected_label`	string

Field	Type
`question`	string
`answer`	string
`route`	string
`agents_run`	array
`predicted_label`	string
`prediction_confidence`	number
`incident_id`	string
`historical_label`	string
`flow_args`	object
`input_quality`	string
`judge_verdict`	null
`judge_overall`	null
`llm_model`	string
`username`	string

Sample request

```
POST /feedback HTTP/1.1
Host: localhost:8000
Authorization: Bearer <token>
Content-Type: application/json

{
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
  "turn_id": "1418ee75519f4518a35e64bb48945262",
  "rating": "down",
  "comment": "Should be Bot",
  "corrected_label": "Bot"
}
```

Sample response

```
HTTP/1.1 201

{
  "id": 1,
  "created_at": "2026-10-11T09:30:25+00:00",
  "updated_at": "2026-10-11T09:30:25+00:00",
  "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
  "turn_id": "1418ee75519f4518a35e64bb48945262",
  "rating": "down",
  "comment": "Should be Bot",
  "corrected_label": "Bot",
  "question": "What attack is incident INC-004393?",
  "answer": "**Assessment**\n\nIncident INC-004393 has been classified as **Infiltration** with 90 %
prediction confidence. The historical database also labels it as Infiltration, providing strong
agreement.\n\n**Why …",
  "route": "classify",
```

```
  "agents_run": [
    "classification"
  ],
  "predicted_label": "Infiltration",
  "prediction_confidence": 0.9,
  "incident_id": "INC-004393",
  "historical_label": "Infiltration",
  "flow_args": {
    "flow_duration": 67917167.0,
    "total_fwd_packets": 20.0,
    "total_bwd_packets": 20.0,
    "total_len_fwd": 3738.0,
    "total_len_bwd": 120.0,
    "flow_packets_s": 0.589,
    "fwd_packets_s": 0.2945,
    "bwd_packets_s": 0.2945,
    "fwd_packet_length_mean": 186.9,
    "bwd_packet_length_mean": 6.0,
    "min_packet_length": 6.0,
    "max_packet_length": 705.0,
    "packet_length_mean": 94.9756,
    "packet_length_std": 181.0974,
    "syn_flag_count": 1.0,
    "ack_flag_count": 1.0,
    "rst_flag_count": 0.0,
    "psh_flag_count": 0.0,
    "fin_flag_count": 0.0,
    "down_up_ratio": 1.0,
    "init_win_bytes_forward": 255.0,
    "init_win_bytes_backward": 1452.0,
    "active_mean": 566477.5,
    "idle_mean": 33100000.0
  },
  "input_quality": "full",
  "judge_verdict": null,
  "judge_overall": null,
  "llm_model": "nvidia:openai/gpt-oss-20b",
  "username": "analyst_demo"
}
```


GET /feedback

List feedback

Stored feedback, newest first.

Property	Value
Method	`GET`
Path	`/feedback`
Authentication	None
Parameters	`rating` (query); `route` (query); `label` (query); `corrected_only` (query); `limit` (query); `offset` (query)

Request format

No request body.

Response format

`200` · application/json

Sample request

```
GET /feedback?limit=5 HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

[
  {
    "id": 1,
    "created_at": "2026-10-11T09:30:25+00:00",
    "updated_at": "2026-10-11T09:30:25+00:00",
    "thread_id": "6ce55595a2714d46ba1d3a5122f2d806",
    "turn_id": "1418ee75519f4518a35e64bb48945262",
    "rating": "down",
    "comment": "Should be Bot",
    "corrected_label": "Bot",
    "question": "What attack is incident INC-004393?",
    "answer": "**Assessment**\n\nIncident INC-004393 has been classified as **Infiltration** with 90 %
prediction confidence. The historical database also labels it as Infiltrat …",
    "route": "classify",
    "agents_run": [
      "classification"
    ],
    "predicted_label": "Infiltration",
    "prediction_confidence": 0.9,
    "incident_id": "INC-004393",
    "historical_label": "Infiltration",
    "flow_args": {
      "flow_duration": 67917167.0,
      "total_fwd_packets": 20.0,
      "total_bwd_packets": 20.0,
      "total_len_fwd": 3738.0,
      "total_len_bwd": 120.0,
      "flow_packets_s": 0.589,
      "fwd_packets_s": 0.2945,
      "bwd_packets_s": 0.2945,
      "fwd_packet_length_mean": 186.9,
      "bwd_packet_length_mean": 6.0,
      "min_packet_length": 6.0,
      "max_packet_length": 705.0,
      "packet_length_mean": 94.9756,
      "packet_length_std": 181.0974,
      "syn_flag_count": 1.0,
      "ack_flag_count": 1.0,
      "rst_flag_count": 0.0,
      "psh_flag_count": 0.0,
      "fin_flag_count": 0.0,
      "down_up_ratio": 1.0,
      "init_win_bytes_forward": 255.0,
      "init_win_bytes_backward": 1452.0,
      "active_mean": 566477.5,
      "idle_mean": 33100000.0
    },
    "input_quality": "full",
    "judge_verdict": null,
    "judge_overall": null,
    "llm_model": "nvidia:openai/gpt-oss-20b",
    "username": "analyst_demo"
  }
]
```

GET /feedback/stats

Feedback statistics

Approval rate (overall and per route), agreement between the LLM judge and analysts, the most common prediction -> correction pairs, and recent comments.

Property	Value
Method	`GET`
Path	`/feedback/stats`
Authentication	None

Request format

No request body.

Response format

`200` · application/json

Field	Type
`total`	number
`up`	number
`down`	number
`approval_rate`	number
`by_route`	object
`judge_agreement`	object
`corrections`	object
`recent_comments`	array

Sample request

```
GET /feedback/stats HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

{
  "total": 1,
  "up": 0,
  "down": 1,
  "approval_rate": 0.0,
  "by_route": {
    "classify": {
      "n": 1,
      "up": 0,
      "approval_rate": 0.0
    }
  },
```

```
  "judge_agreement": {
    "n": 0,
    "pass_up": 0,
    "pass_down": 0,
    "not_pass_up": 0,
    "not_pass_down": 0,
    "agreement_rate": null
  },
  "corrections": {
    "total": 1,
    "top_pairs": [
      {
        "predicted": "Infiltration",
        "corrected": "Bot",
        "count": 1
      }
    ]
  },
  "recent_comments": [
    {
      "rating": "down",
      "route": "classify",
      "comment": "Should be Bot",
      "created_at": "2026-10-11T09:30:25+00:00"
    }
  ]
}
```


GET /feedback/export.csv

Export label corrections

Corrected flows for retraining: 24 features, corrected label (majority of analysts), source incident, votes. Corrections without usable features are left out.

Property	Value
Method	`GET`
Path	`/feedback/export.csv`
Authentication	None

Request format

No request body.

Response format

`200` · text/csv

Sample request

```
GET /feedback/export.csv HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

flow_duration,total_fwd_packets,total_bwd_packets,total_len_fwd,total_len_bwd,flow_packets_s,… (28 columns)
67917167.0,20.0,20.0,3738.0,120.0,0.588952716,…
```

6. Machine-learning models

POST /ml/classify

Attack type of a flow

Predicted attack type (15 classes incl. BENIGN), confidence and alternatives.

Property	Value
Method	`POST`
Path	`/ml/classify`
Authentication	None

Request format

A flow (see Flow input): any subset of the 24 metrics, or an `incident_id`.

Response format

`200` · application/json

Field	Type
`incident`	object
`status`	string
`input`	object
`attack_type`	object
`overall_confidence`	string

Sample request

```
POST /ml/classify HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "incident_id": "INC-000035"
}
```

Sample response

```
HTTP/1.1 200

{
  "incident": {
    "incident_id": "INC-000035",
    "historical_label": "SSH-Patator",
    "dst_port": 22
  },
  "status": "ok",
  "input": {
    "input_quality": "full",
    "known_metrics": "24/24",
    "derived": [],
```

```
    "filled_with_normal_values": [],
    "warnings": []
  },
  "attack_type": {
    "prediction": "SSH-Patator",
    "confidence": 1.0,
    "alternatives": []
  },
  "overall_confidence": "high"
}
```

POST /ml/risk

Intrusion risk and anomaly of a flow

Calibrated intrusion probability, risk level, and anomaly level vs normal traffic.

Property	Value
Method	`POST`
Path	`/ml/risk`
Authentication	None

Request format

A flow (see Flow input): any subset of the 24 metrics, or an `incident_id`.

Response format

`200` · application/json

Field	Type
`status`	string
`input`	object
`risk`	object
`overall_confidence`	string
`anomaly`	object

Sample request

```
POST /ml/risk HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "flow_duration": "2 s",
  "total_fwd_packets": 10,
  "total_bwd_packets": 8,
  "total_len_fwd": "1.2 KB",
  "total_len_bwd": "4 KB"
}
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "input": {
    "input_quality": "reduced",
    "known_metrics": "12/24",
    "derived": [
      "flow_packets_s",
      "fwd_packets_s",
      "… 5 more"
    ],
    "filled_with_normal_values": [
      "min_packet_length",
      "max_packet_length",
      "… 10 more"
    ],
    "warnings": [
      "packet_length_mean was approximated from total bytes and packets",
      "12 feature(s) not provided were filled with normal-traffic medians"
    ]
  },
  "risk": {
    "probability": 0.0048,
    "is_intrusion": false,
    "level": "Low"
  },
  "overall_confidence": "medium",
  "anomaly": {
    "score": 0.5286,
    "benign_percentile": 81.8,
    "level": "Low",
    "most_unusual_features": []
  }
}
```

POST /ml/explain

Full analysis with SHAP reasons

Attack type, risk, anomaly, priority, and the metrics that drove each prediction (SHAP).

Property	Value
Method	`POST`
Path	`/ml/explain`
Authentication	None

Request format

A flow (see Flow input): any subset of the 24 metrics, or an `incident_id`.

Response format

`200` · application/json

Field	Type
`incident`	object

Field	Type
`status`	string
`input`	object
`attack_type`	object
`risk`	object
`anomaly`	object
`overall_confidence`	string
`explanation`	object
`priority`	object

Sample request

```
POST /ml/explain HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "incident_id": "INC-000035"
}
```

Sample response

```
HTTP/1.1 200

{
  "incident": {
    "incident_id": "INC-000035",
    "historical_label": "SSH-Patator",
    "dst_port": 22
  },
  "status": "ok",
  "input": {
    "input_quality": "full",
    "known_metrics": "24/24",
    "derived": [],
    "filled_with_normal_values": [],
    "warnings": []
  },
  "attack_type": {
    "prediction": "SSH-Patator",
    "confidence": 1.0,
    "alternatives": []
  },
  "risk": {
    "probability": 1.0,
    "is_intrusion": true,
    "level": "Critical"
  },
  "anomaly": {
    "score": 0.504,
    "benign_percentile": 71.8,
    "level": "Low",
    "most_unusual_features": [
      {
        "feature": "total_bwd_packets",
```

```
        "value": 32.0,
        "benign_percentile": 96.7,
        "text": "Backward packets = 32 packets (higher than 97% of normal traffic)"
      }
    ]
  },
  "overall_confidence": "high",
  "explanation": {
    "attack_type_reasons": [
      {
        "feature": "total_bwd_packets",
        "value": 32.0,
        "shap": 0.1875,
        "effect": "toward SSH-Patator",
        "benign_percentile": 96.7,
        "text": "Backward packets = 32 packets (higher than 97% of normal traffic)"
      },
      {
        "feature": "total_fwd_packets",
        "value": 21.0,
        "shap": 0.1619,
        "effect": "toward SSH-Patator",
        "benign_percentile": 93.5,
        "text": "Forward packets = 21 packets (higher than 94% of normal traffic)"
      },
      "… 1 more"
    ],
    "risk_reasons": [
      {
        "feature": "init_win_bytes_forward",
        "value": 29200.0,
        "shap": 4.3939,
        "effect": "raises risk",
        "benign_percentile": 88.6,
        "text": "Initial TCP window (forward) = 29,200 bytes (higher than 89% of normal traffic)"
      },
      {
        "feature": "init_win_bytes_backward",
        "value": 247.0,
        "shap": 2.2831,
        "effect": "raises risk",
        "benign_percentile": 71.5,
        "text": "Initial TCP window (backward) = 247 bytes (higher than 72% of normal traffic)"
      },
      "… 1 more"
    ],
    "summary": "Classified as SSH-Patator (100% confidence). Intrusion risk 100% (Critical). Anomaly level
Low. Main reasons: Backward packets = 32 packets (higher than 97% of normal traffic); Forward packets = 21
packets (higher than 94% of normal traffic); Forward bytes = 2 …"
  },
  "priority": {
    "priority": "P2",
    "score": 66.0,
    "label": "High",
    "action": "Investigate today",
    "reason": "SSH-Patator (100%, severity Medium), intrusion risk 100%, anomaly Low",
    "components": {
      "risk": 1.0,
      "severity": 0.6,
      "anomaly": 0.0
    },
    "overrides_applied": [],
    "confidence_note": null
  }
}
```

POST /ml/prioritize

Priority of one flow

P1 (Critical) .. P4 (Low) with score, reason and rule overrides.

Property	Value
Method	`POST`
Path	`/ml/prioritize`
Authentication	None

Request format

A flow (see Flow input): any subset of the 24 metrics, or an `incident_id`.

Response format

`200` · application/json

Field	Type
`incident`	object
`status`	string
`input`	object
`attack_type`	object
`risk`	object
`anomaly`	object
`overall_confidence`	string
`priority`	object

Sample request

```
POST /ml/prioritize HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "incident_id": "INC-002545"
}
```

Sample response

```
HTTP/1.1 200

{
  "incident": {
    "incident_id": "INC-002545",
    "historical_label": "Heartbleed",
    "dst_port": 444
  },
  "status": "ok",
  "input": {
    "input_quality": "full",
```

```
    "known_metrics": "24/24",
    "derived": [],
    "filled_with_normal_values": [],
    "warnings": []
  },
  "attack_type": {
    "prediction": "Heartbleed",
    "confidence": 1.0,
    "alternatives": []
  },
  "risk": {
    "probability": 1.0,
    "is_intrusion": true,
    "level": "Critical"
  },
  "anomaly": {
    "score": 0.6566,
    "benign_percentile": 99.9,
    "level": "High",
    "most_unusual_features": [
      {
        "feature": "bwd_packet_length_mean",
        "value": 3808.3789,
        "benign_percentile": 100.1,
        "text": "Mean backward packet size = 3,808.38 bytes (higher than 99.9% of normal traffic)"
      },
      {
        "feature": "total_fwd_packets",
        "value": 2801.0,
        "benign_percentile": 100.0,
        "text": "Forward packets = 2,801 packets (higher than 99.9% of normal traffic)"
      },
      "… 1 more"
    ]
  },
  "overall_confidence": "high",
  "priority": {
    "priority": "P1",
    "score": 99.8,
    "label": "Critical",
    "action": "Investigate now",
    "reason": "Heartbleed (100%, severity Critical), intrusion risk 100%, anomaly High",
    "components": {
      "risk": 1.0,
      "severity": 1.0,
      "anomaly": 0.99
    },
    "overrides_applied": [],
    "confidence_note": null
  }
}
```


POST /ml/prioritize/batch

Rank many flows / incidents

Rank flows and/or incident IDs by urgency; returns the top_n and counts per priority.

Property	Value
Method	`POST`
Path	`/ml/prioritize/batch`
Authentication	None

Request format

Field	Type	Required	Description
`flows`	array of object	no	(max items 1000)
`incident_ids`	array of string	no	(max items 1000)
`top_n`	integer	no	How many of the most urgent flows to return. (≥ 1.0, ≤ 1000.0, default 10)

Response format

`200` · application/json

Field	Type
`status`	string
`total_flows`	number
`priority_counts`	object
`top`	array

Sample request

```
POST /ml/prioritize/batch HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "incident_ids": [
    "INC-000001",
    "INC-000035",
    "INC-002545",
    "INC-000005"
  ],
  "top_n": 3
}
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "total_flows": 4,
  "priority_counts": {
    "P1": 2,
    "P2": 1,
    "P3": 1
  },
  "top": [
    {
      "incident_id": "INC-002545",
      "historical_label": "Heartbleed",
      "priority": "P1",
      "score": 99.8,
      "attack_type": "Heartbleed",
      "confidence": 1.0,
      "risk_probability": 1.0,
      "anomaly_level": "High",
      "reason": "Heartbleed (100%, severity Critical), intrusion risk 100%, anomaly High",
```

```
      "overrides": [],
      "input_quality": "full"
    },
    {
      "incident_id": "INC-000001",
      "historical_label": "DoS Hulk",
      "priority": "P1",
      "score": 85.6,
      "attack_type": "DoS Hulk",
      "confidence": 1.0,
      "risk_probability": 1.0,
      "anomaly_level": "Medium",
      "reason": "DoS Hulk (100%, severity High), intrusion risk 100%, anomaly Medium",
      "overrides": [],
      "input_quality": "full"
    },
    "… 1 more"
  ]
}
```


POST /ml/prioritize/csv

Rank flows from a CSV file

Upload a CSV (max 5 MB, 1000 rows). Each row is a flow; `flow_index` in the result is the row number.

Property	Value
Method	`POST`
Path	`/ml/prioritize/csv`
Authentication	None
Parameters	`top_n` (query)

Request format

`multipart/form-data` with a `file` field: a CSV of flows (≤ 5 MB, ≤ 1,000 rows). Columns are matched by metric name, alias or raw CIC-IDS2017 header; an `incident_id` column is also accepted.

Response format

`200` · application/json

Field	Type
`status`	string
`total_flows`	number
`priority_counts`	object
`top`	array

Sample request

```
POST /ml/prioritize/csv?top_n=3 HTTP/1.1
Host: localhost:8000
Content-Type: multipart/form-data

file = flows.csv:
incident_id
```

```
INC-000001
INC-000035
INC-002545
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "total_flows": 3,
  "priority_counts": {
    "P1": 2,
    "P2": 1
  },
  "top": [
    {
      "incident_id": "INC-002545",
      "historical_label": "Heartbleed",
      "priority": "P1",
      "score": 99.8,
      "attack_type": "Heartbleed",
      "confidence": 1.0,
      "risk_probability": 1.0,
      "anomaly_level": "High",
      "reason": "Heartbleed (100%, severity Critical), intrusion risk 100%, anomaly High",
      "overrides": [],
      "input_quality": "full"
    },
    {
      "incident_id": "INC-000001",
      "historical_label": "DoS Hulk",
      "priority": "P1",
      "score": 85.6,
      "attack_type": "DoS Hulk",
      "confidence": 1.0,
      "risk_probability": 1.0,
      "anomaly_level": "Medium",
      "reason": "DoS Hulk (100%, severity High), intrusion risk 100%, anomaly Medium",
      "overrides": [],
      "input_quality": "full"
    },
    "… 1 more"
  ]
}
```

7. Incident search

POST /search

Find similar historical incidents

Hybrid keyword + semantic search over ~97,000 incidents, with optional attack-type and port filters and numeric conditions in the query ("more than 10000 packets per second"). Retrieval only: no generated answer.

The first search waits for the indexes if they are still loading (see GET /health).

Property	Value
Method	`POST`
Path	`/search`
Authentication	None

Request format

Field	Type	Required	Description
`query`	string	yes	Description of the traffic or incident. (min length 1, max length 1000)
`attack_type`	string	no	Only incidents of this attack type.
`dst_port`	integer	no	Only incidents on this port.
`top_k`	integer	no	(≥ 1.0, ≤ 20.0, default 5)

Response format

`200` · application/json

Field	Type
`status`	string
`query`	string
`filters_applied`	object
`filters_inferred_from_query`	object
`numeric_conditions`	array
`candidates_after_filters`	number
`evidence_available`	boolean
`incidents`	array

Sample request

```
POST /search HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
```

```
  "query": "SSH brute force with many short connections",
  "attack_type": "SSH-Patator",
  "top_k": 3
}
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "query": "SSH brute force with many short connections",
  "filters_applied": {
    "label": "SSH-Patator"
  },
  "filters_inferred_from_query": {},
  "numeric_conditions": [],
  "candidates_after_filters": 3149,
  "evidence_available": true,
  "incidents": [
    {
      "incident_id": "INC-002563",
      "attack_type": "SSH-Patator",
      "dst_port": 22,
      "relevance_score": 0.0164,
      "summary": "Attack: SSH-Patator. Destination port: 22. Flow duration: 50 microseconds. Forward packets:
1. Backward packets: 1. Forward bytes: 0. Backward bytes: 0. Flow packets per second: 40000. Forward packets
per second: 20000. …"
    },
    {
      "incident_id": "INC-079784",
      "attack_type": "SSH-Patator",
      "dst_port": 22,
      "relevance_score": 0.0164,
      "summary": "Attack: SSH-Patator. Destination port: 22. Flow duration: 2138428 microseconds. Forward
packets: 20. Backward packets: 18. Forward bytes: 2296. Backward bytes: 3069. Flow packets per second:
17.7701. Forward packets per …"
    },
    "… 1 more"
  ]
}
```

GET /incidents/{incident_id}

Look up a historical incident

Stored metrics, destination port and historical label of an incident (e.g. INC-000035).

Property	Value
Method	`GET`
Path	`/incidents/{incident_id}`
Authentication	None
Parameters	`incident_id` (path, required)

Request format

No request body.

Response format

`200` · application/json

Field	Type
`status`	string
`incident_id`	string
`historical_label`	string
`dst_port`	number
`input`	object
`metrics`	object

Sample request

```
GET /incidents/INC-000035 HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "incident_id": "INC-000035",
  "historical_label": "SSH-Patator",
  "dst_port": 22,
  "input": {
    "input_quality": "full",
    "known_metrics": "24/24",
    "derived": [],
    "filled_with_normal_values": [],
    "warnings": []
  },
  "metrics": {
    "Flow duration": 13903754.0,
    "Forward packets": 21.0,
    "Backward packets": 32.0,
    "Forward bytes": 2008.0,
    "Backward bytes": 2745.0,
    "Flow packet rate": 3.8119,
    "Forward packet rate": 1.5104,
    "Backward packet rate": 2.3015,
    "Mean forward packet size": 95.619,
    "Mean backward packet size": 85.7812,
    "Minimum packet size": 0.0,
    "Maximum packet size": 976.0,
    "Mean packet size": 88.0185,
    "Packet size std. dev.": 189.5903,
    "SYN flag": 0.0,
    "ACK flag": 0.0,
    "RST flag": 0.0,
    "PSH flag": 1.0,
    "FIN flag": 0.0,
    "Download/upload ratio": 1.0,
    "Initial TCP window (forward)": 29200.0,
    "Initial TCP window (backward)": 247.0,
    "Mean active time": 0.0,
    "Mean idle time": 0.0
```

```
  }
}
```

GET /attack-types

List attack types

The 15 classes the models know (incl. BENIGN) with their severity.

Property	Value
Method	`GET`
Path	`/attack-types`
Authentication	None

Request format

No request body.

Response format

`200` · application/json

Field	Type
`status`	string
`attack_types`	array

Sample request

```
GET /attack-types HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "attack_types": [
    {
      "attack_type": "BENIGN",
      "severity": "None"
    },
    {
      "attack_type": "Bot",
      "severity": "High"
    },
    {
      "attack_type": "DDoS",
      "severity": "High"
    },
    "… 12 more"
  ]
}
```

GET /attack-types/{name}

Profile of an attack type

Typical traffic of an attack type: which metrics differ from normal traffic, with medians.

Property	Value
Method	`GET`
Path	`/attack-types/{name}`
Authentication	None
Parameters	`name` (path, required)

Request format

No request body.

Response format

`200` · application/json

Field	Type
`status`	string
`attack_type`	string
`severity`	string
`summary`	string
`top_features`	array

Sample request

```
GET /attack-types/DDoS HTTP/1.1
Host: localhost:8000
```

Sample response

```
HTTP/1.1 200

{
  "status": "ok",
  "attack_type": "DDoS",
  "severity": "High",
  "summary": "Initial TCP window (backward) typically higher than normal (median 229 vs 0); Backward bytes typically higher than normal (median 11,601 vs 163); Mean backward packet size typically higher than normal (median 1,934.50 vs 95); Forward bytes typically lower than …",
  "top_features": [
    {
      "feature": "Initial TCP window (backward)",
      "direction_vs_normal": "higher",
      "attack_median": 229.0,
      "normal_median": 0.0,
      "unit": "bytes",
      "importance_share": 0.1043
    },
    {
      "feature": "Backward bytes",
```

```
      "direction_vs_normal": "higher",
      "attack_median": 11601.0,
      "normal_median": 163.0,
      "unit": "bytes",
      "importance_share": 0.1042
    },
    "… 3 more"
  ]
}
```